Andrea Panebianco, Ph.D.

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h-index: 7 | 119 citations (Google Scholar) · 101 citations (Scopus) — as of May 2026

Profile

Postdoctoral Researcher in underwater acoustic communications and wireless networking, with a research background spanning digital communications, signal/data processing, channel modelling, adaptive protocols, and learning-based resource allocation for the Internet of Underwater Things (IoUT). My work combines model-based and data-driven methods to predict underwater channel conditions, support physical-layer-aware link adaptation, optimize modulation, power control, feedback scheduling, and routing, and improve reliability, energy efficiency, and resilience in constrained maritime networks. I have worked with real underwater acoustic experimental traces from the NATO CMRE LOON testbed and have published on underwater channel prediction, adaptive modulation, OFDM carrier selection, learning-based network adaptation, and jamming-resilient underwater routing. My research also includes protocol-level design, simulation-based evaluation, and integration of heterogeneous underwater sensing and communication systems, including acoustic/radio IoUT architectures for maritime environments.

Education

- **Ph.D. in Information and Communication Technologies** — Nov. 2022 – Dec. 2025
  Department of Engineering, University of Palermo, Palermo, Italy
  Degree awarded: 17 Dec. 2025
  Thesis: "Learning and Adaptation for Efficient and Resilient Internet of Underwater Things Communications"
  Developed model-based and learning-based strategies for underwater acoustic digital communications and Internet of Underwater Things (IoUT) networks, with emphasis on channel prediction, physical-layer-aware link adaptation, adaptive modulation, power control, feedback scheduling, energy-efficient resource allocation, and routing resilience. The work combined signal/data analysis, online learning, multi-armed bandits, and reinforcement learning, and was validated through simulations under realistic underwater channel, latency, and energy constraints.
- **M.Sc. in Telecommunications Engineering** — Oct. 2019 – Oct. 2022
  Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
  Degree awarded: 26 Oct. 2022
  Thesis: "Characterization of Underwater Communication Channels in Shallow Water Environments: From On-Field Measurements to Markov-Based Models"
  Developed and validated finite-state Markov models for shallow-water underwater acoustic channels using real-world experimental traces from the NATO CMRE LOON testbed. The work focused on physical-layer channel characterization, underwater acoustic trace processing, channel impulse response analysis, SNR estimation, channel-quality assessment, propagation-induced variability, and MATLAB-based model comparison.
- **B.Sc. in Electronic Engineering** — Sep. 2015 – Mar. 2020
  Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
  Degree awarded: 20 Mar. 2020
  Thesis: "Implementation of a Haptic Interface for Tactile Internet Scenarios in 5G Networks"
  Designed and implemented a Hapkit-based haptic interface for Tactile Internet applications in 5G-oriented networks, integrating Arduino-based device control with a Mininet/SDN emulation environment and evaluating packet-level latency under different network-delay and routing conditions.

Professional Experience

- **Postdoctoral Researcher** — Jan. 2026 – Present
  Department of Electrical and Computer Engineering, Auburn University, AL, USA
  Conducting research on adaptive communication systems for resource-constrained wireless networks and semantic/goal-oriented communications. The work focuses on cross-layer optimization, online learning, network control, and learning-based algorithms for communication, routing, scheduling, and resource allocation in dynamic environments. Contributing to research proposal and grant preparation with the project supervisor, including technical writing, problem formulation, research planning, and alignment of proposed activities

with project objectives.

- **Visiting Ph.D. Student** Jan. 2025 – Jul. 2025
  Department of Electrical and Computer Engineering, Auburn University, AL, USA
  Developed learning-based and age-aware communication mechanisms for underwater acoustic networks, with emphasis on limited feedback, adaptive link control, modulation and power selection, and bandit-based resource allocation. The work supported the design of lightweight online algorithms for improving reliability, energy efficiency, and adaptability in constrained underwater communication systems.

- **Teaching Assistant** Mar. 2023 – Jun. 2023; Mar. 2024 – Jun. 2024
  Course: IoT and Big Data Sensing, Compression, and Communication
  Department of Electrical, Electronic and Computer Engineering, University of Catania, Italy
  Supported teaching activities, laboratory sessions, and student supervision on IoT systems, sensing architectures, data compression, and communication technologies, with focus on practical implementation and performance evaluation.

- **Research Intern** Sep. 2020 – Jun. 2021
  Department of Electrical, Electronic and Computer Engineering, University of Catania, Italy
  Contributed to research and development activities on underwater communication systems and IoT-based monitoring architectures, including LoRa connectivity, MQTT-based data exchange, digital-twin-oriented infrastructure monitoring, web dashboards, and integrated sensing/communication prototypes.

Technical Skills

- **Programming, Simulation, and Tools:** MATLAB, Python, Java, Arduino, DESERT Underwater, Mininet, Wireshark, LaTeX; algorithm prototyping, underwater network simulation, protocol-level evaluation, and experimental data analysis.
- **Underwater and Digital Communications:** underwater acoustic communications, Internet of Underwater Things (IoUT), adaptive modulation, OFDM-based carrier selection, power control, feedback scheduling, routing, protocol design, acoustic/radio system integration, and scalable underwater networking architectures.
- **Signal and Data Processing:** underwater acoustic trace processing, physical-layer performance analysis, channel impulse response analysis, SNR estimation, channel-quality characterization, propagation-aware link modelling, data-driven channel prediction, Markov and Hidden Markov modelling, and Kalman filtering.
- **Learning, Optimization, and Networking:** multi-armed bandits, contextual and combinatorial bandits, reinforcement learning, deep reinforcement learning, online learning, Age-of-Information-aware optimization, cross-layer resource allocation, energy-aware networking, and jamming-resilient communications.
- **Research Development and Technical Writing:** scientific manuscript preparation, technical reporting, technical presentations, technical contributions to research proposal and grant preparation, problem formulation, research planning, literature review, and alignment with sponsor-driven objectives.

Experimental and Field Trial Experience

- Participated in nearshore shallow-water marine field tests of an integrated acoustic/LoRa Internet of Underwater Things (IoUT) system at La Playa, Catania, Italy, involving the deployment and testing of underwater acoustic modems, sensing devices, control boards, LoRa connectivity, and remote monitoring interfaces.
- Contributed to the experimental validation of underwater multimedia and control-data transmission, including system configuration, data acquisition, performance evaluation, and analysis of transmission delay, packet delivery, and achievable bit rate under real shallow-water conditions.
- Worked with real underwater acoustic experimental traces from the NATO CMRE LOON testbed for shallow-water channel characterization, including channel impulse response analysis, SNR estimation, channel-quality assessment, and Markov-based channel modelling.

Publications

Unless otherwise indicated, author order follows the convention of the corresponding research community. Equal-contribution authorship and contribution-based ordering are explicitly noted.

Peer-Reviewed Journal Articles

- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R. (2025). Bandits Under the Waves: A Fully-Distributed Multi-Armed Bandit Framework for Modulation Adaptation in the Internet of Underwater Things. *IEEE Transactions on Network and Service Management*, 23, 639–653.

- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R. (2025). AMUSE: A Multi-Armed Bandit Framework for Energy-Efficient Modulation Adaptation in Underwater Acoustic Networks. *IEEE Open Journal of the Communications Society*, 6, 2766–2779.
- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, Qi, Z., & Pompili, D. (2024). Adaptive versus Predictive Techniques in Underwater Acoustic Communication Networks. *Computer Networks*, 252, 110679. *[Authors 1–5 listed alphabetically]*
- Busacca, F., Galluccio, L., Palazzo, S., & **Panebianco, A.** (2024). A Comparative Analysis of Predictive Channel Models for Real Shallow-Water Environments. *Computer Networks*, 250, 110557.
Research conducted using real-world experimental acoustic traces provided by NATO CMRE.
- Mertens, J. S., **Panebianco, A.**, Surudhi, A., Prabagarane, N., & Galluccio, L. (2023). Network Intelligence vs. Jamming in Underwater Networks: How Learning Can Cope with Misbehavior. *Frontiers in Communications and Networks*, 4, 1179626. *[Listed by contribution]*
- Brincat, A. A., Busacca, F., Galluccio, L., Mertens, J. S., Musumeci, A., Palazzo, S., & **Panebianco, A.** (2022). An Integrated Acoustic/LoRa System for Transmission of Multimedia Sensor Data over an Internet of Underwater Things. *Computer Communications*, 192, 132–142.

Peer-Reviewed Conference Papers

- Busacca, F., **Panebianco, A.**, Raftopoulos, R., & Scuderi, D. (2026, June). VISTA: Velocity-Informed Smart Transmission Adaptation in High-Mobility IoT. In *Proc. IEEE/IFIP International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks Workshops (WiOpt Workshops 2026)* (pp. 9–15).
- Busacca, F., **Panebianco, A.**, & Sun, Y. (2025, December). Adaptive Underwater Acoustic Communications with Limited Feedback: An AoI-Aware Hierarchical Bandit Approach. In *Proc. IEEE Global Communications Conference (GLOBECOM 2025)* (pp. 2426–2431).
- Busacca, F., Galluccio, L., **Panebianco, A.**, Palazzo, S., Raftopoulos, R., & Scuderi, D. (2025, September). MERMAID: A Multi-Armed Bandit Approach for Optimal Carrier Selection in OFDM-Based Underwater Acoustic Networks. In *Proc. IEEE International Symposium on Personal, Indoor and Mobile Radio Communications Workshops (PIMRC Workshops 2025)* (pp. 1–6). *[Listed by contribution]*
- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R. (2025, June). POSEIDON: A Multi-Armed Bandit Framework for Power and Modulation Scheme Adaptation in Underwater Acoustic Networks. In *Proc. IEEE International Conference on Communications Workshops (ICC Workshops 2025)* (pp. 2045–2050).
- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R. (2025, March). A Distributed Multi-Armed Bandit Approach for Modulation Adaptation in Underwater Networks. In *Proc. IEEE Wireless Communications and Networking Conference (WCNC 2025)* (pp. 1–6).
- Scarvaglieri, A., **Panebianco, A.**, & Busacca, F. (2024, December). MAGELLAN: A Distributed MAB-Based Algorithm for Energy-Fair and Reliable Routing in Multi-Hop LoRa Networks. In *Proc. IEEE Global Communications Conference (GLOBECOM 2024)* (pp. 2250–2255). *[Listed by contribution]*
- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Raftopoulos, R. (2024, June). Adaptive Modulation in Underwater Acoustic Networks (AMUSE): A Multi-Armed Bandit Approach. In *Proc. IEEE International Conference on Communications (ICC 2024)* (pp. 2336–2341).
- Busacca, F., Galluccio, L., Palazzo, S., **Panebianco, A.**, & Scarvaglieri, A. (2024, June). Balancing Optimization for Underwater Network Cost Effectiveness (BOUNCE): A Multi-Armed Bandit Solution. In *Proc. IEEE International Conference on Communications Workshops (ICC Workshops 2024)* (pp. 1340–1345).
- Busacca, F., Galluccio, L., Palazzo, S., & **Panebianco, A.** (2023, May). Underwater Acoustic Channel Models for SNR Prediction in a Real Shallow-Water Environment. In *Proc. IEEE International Conference on Communications Workshops (ICC Workshops 2023)* (pp. 721–726).
Research conducted using real-world experimental acoustic traces provided by NATO CMRE.

Patent Applications

- Sun, Y., Chakraborty, S., McDowell, C. D., & **Panebianco, A.** (2026). *Significance-Driven Semantic Communication.* U.S. Provisional Patent Application No. 64/065,591, filed May 14, 2026.
- Sun, Y., Chakraborty, S., Chamoun, S. G., McDowell, C. D., **Panebianco, A.**, & Ross, T. P. (2025). *Spectrum Efficiency of Semantics and Goal Oriented Communications.* U.S. Provisional Patent Application

No. 63/875,119, filed Sep. 3, 2025.

Research Projects

- **Development of NextG Wireless Testbeds for Semantic and Goal-Oriented Communication** Jan. 2026 – Present
  Department of Electrical and Computer Engineering, Auburn University, Auburn, AL, USA
  Development and integration of an end-to-end wireless testbed for semantic and goal-oriented communications, with emphasis on protocol implementation, experimental validation, resource-constrained networking, and mission-critical IoT applications.

- **AQUASMARTT - Data acquisition with underwater and above-water sustainable wireless nodes for monitoring how climatic changes affects natural disasters and biodiversity** Nov. 2022 – Dec. 2025
  Department of Engineering, University of Palermo, Palermo, Italy
  Research and development of intelligent networking components for coastal sensing and underwater acoustic networks. The project included channel-quality prediction, adaptive modulation and power control, Age-of-Information-aware feedback scheduling, energy-efficient resource allocation, DESERT-based simulation, and protocol-level evaluation for Internet of Underwater Things (IoUT) scenarios. The work also involved integration aspects for heterogeneous underwater sensing and communication systems.

- **4Frailty - Smart sensors, infrastructures, and management models for the safety of frail patients** Jul. 2023 – Oct. 2023
  Department of Engineering, University of Palermo, Palermo, Italy
  Development of an IoT-based remote-care monitoring platform integrating wearable sensors, Android interfaces, MQTT-based data exchange, LoRa connectivity, and secure data storage.

- **Secure and Efficient Wireless Communications - Intelligent Communication Algorithms and Integrated System Development for Underwater Wireless Communication Environments** Sep. 2020 – May 2021
  Intelligent Communication Algorithms and Integrated System Development for Underwater Wireless Communication Environments
  Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
  Design, integration, and field testing of intelligent communication and monitoring architectures for underwater wireless environments. The work included nearshore shallow-water marine tests of an integrated acoustic/LoRa IoUT system at La Playa, Catania, underwater acoustic modem operation, IoT-based sensing, MQTT-based data exchange, LoRa connectivity, digital-twin modules, web dashboards, and simulated wake-up radio mechanisms.

Honors and Awards

- **Francesco Carassa Award** Sep. 2023
  Italian National Group on Telecommunications and Information Theory (GTTI) Annual Meeting 2023, Rome, Italy
  Awarded for the poster: *"Characterization of Underwater Acoustic Channel Models for SNR Prediction in Real Shallow-Water Environments."*
  Research conducted using real-world experimental acoustic traces from NATO CMRE.

- **Student Award "TLC for a Better World" – Innovative Prototype Category** Dec. 2019
  TelcomDay 2019, Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy
  Awarded for the project: *"Implementation of a Haptic Interface for Tactile Internet Scenarios in 5G Networks."*

Professional Activities and Service

- **Appointed Organizing Committee Member** Nov. 2026
  Third Workshop on Artificial Intelligence of Things (AIoT 2026)
  Held in conjunction with the 27th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc 2026), Tokyo, Japan

- **Technical Program Committee Member** Oct. 2026
  IEEE Military Communications Conference (MILCOM 2026) — Track 1: Waveforms and Signal Processing, Washington, DC, USA

IEEE Military Communications Conference (MILCOM 2026) — Workshop (WS7): Quality, Value, and Age of Information for Tactical Networks, Washington, DC, USA

- **Finance and Registration Chair** Jun. 2026
  24th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt 2026), hosted by The Ohio State University, Columbus, OH, USA
- **Technical Program Committee Chair** Jun. 2026
  Third Workshop on Modeling and Optimization for Semantic Communications (MOSC 2026)
  Held in conjunction with the 24th International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt 2026), hosted by The Ohio State University, Columbus, OH, USA

Invited Talks and Seminars

- **A Multi-Armed Bandit Approach in Underwater Networks** Feb. 2024
  Winter School on Underwater Network Simulations and Experimentation (UNWiS), Department of Information Engineering, University of Padova, Padova, Italy
- **Design and Experimentation of an Integrated Acoustic/LoRa System for Transmission of Multimedia Sensor Data over an Internet of Underwater Things** May 2021
  Seminar on Development and Applications of Innovative Materials and Processes for the Diagnostics and Restoration of Cultural Heritage, Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy

Mentorship

- **Co-Supervisor of Master's Thesis** Jan. 2025 – Oct. 2025
  Student: Dario Scuderi
  Thesis: *"Smart Carrier Selection in OFDM-Based Acoustic Networks for the Internet of Underwater Things"*
  Master's Degree in Communications Engineering, Department of Electrical, Electronic and Computer Engineering, University of Catania, Catania, Italy

Certifications

- **Qualification for the Profession of Information Engineer** Nov. 2023
  University of Catania, Italy
- **BI Level C2 Certificate in ESOL International (CEFR C2)** May 2022
  British Institutes

Additional Information

- **Nationality:** Italian.
- **Languages:** Italian, native; English, C2 CEFR.

Professional Membership

- IEEE Member
- IEEE Young Professionals Member
- IEEE Communications Society Member